Data Project Part 1

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.1     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(readr)
library(ggplot2)
```

1.[2 marks] The first part of our PPDAC framework is to identify the problem you are addressing with these data. State the question you are trying to answer and let us know what type of question this is in terms of the PPDAC framework. A question statement should be as specific as possible. For example:Do students who regularly get 8 hours of sleep have fewer visits to the health center? This question is an example of an etiologic or causal question.

How does the age-adjusted death rate for heart disease vary across different states in the United States in 2017, and are there any observable patterns or trends based on geographic regions (e.g., Northeast, South, Midwest, West)?

2. [2 marks] Why is this question interesting or important? You could talk here about how existing data/studies suggest this might be important, how the findings might make an impact, how the findings might be used, or why you are personally interested in this question.

Heart disease is one of the leading causes of death in the United States, and understanding how its mortality rates vary across states can help identify regions that may need targeted public health interventions. If certain states or regions consistently show higher death rates, it could indicate disparities in healthcare access, lifestyle factors, or environmental conditions that need to be addressed. The findings from this analysis could inform policymakers and healthcare providers about where to allocate resources, such as funding for prevention programs, healthcare infrastructure, or public awareness campaigns. For example, if the South has higher heart disease death rates, it might prompt initiatives to improve diet, exercise, and smoking cessation programs in that region. Observing geographic patterns in heart disease mortality could reveal insights into the role of regional factors such as diet (e.g., prevalence of high-fat or high-sodium diets), socioeconomic status, or cultural practices, leading to more tailored and effective public health strategies.

3. [2 marks] What is the target population for your project? Why was this target chosen? (i.e., what was your rationale for wanting to answer this question in this specific population?)

The target population for this project is the general U.S. population across different states. This target population was chosen because understanding leading causes of death across the country can help inform public health policies, allocate healthcare resources effectively, and identify trends in mortality rates over time.

4. [2 marks] What is the sampling frame used to collect the data you are using? It may be helpful here to read any protocol papers, trial registration records, '.Readme' files or documentation that are associated with your dataset. If you have trouble identifying how the records/individuals were sampled, confirm with your supporting GSI that your dataset will be usable for the purposes of the class. Describe why you think this sampling strategy is appropriate for your question. To what group(s) would you feel comfortable generalizing the findings of your study and why?

The dataset categorizes deaths by state, cause of death, and year, meaning that it was collected by stratifying the population based on these factors. It contains representative samples that ensures coverage across different regions and reasons. This strategy ensures proportional representation of each state, making regional comparisons possible, comprehensively records major causes of death, and minimizes bias. Additionally, annual trend analysis is highly valuable for public health research. The dataset contains mortality data categorized by state, making it suitable for analyzing state-level trends.

5. [2 marks] Write a brief description (1-4 sentences) of the source and contents of your dataset. Provide a URL to the original data source if applicable. If not (e.g., the data came from your internship),provide 1-2 sentences saying where the data came from. If you completed a web form to access the data and selected a subset, describe these steps (including any options you selected) and the date you accessed the data.

The dataset is sourced from National Center for Health Statistics (NCHS) that documents NCHS Leading Causes of Death in the United States. It contains information on the mortality data for various causes of death across different US states and years, including variables such as number of deaths, age-adjusted death rates, and cause classifications. The data were collected to evaluate changes in mortality rates over time across different states and demographic groups. I found this dataset from a website (https://catalog.data.gov/dataset/nchs-leading-causes-of-death-united-states/resource/9096aa3c-0d4b-42f1-bb01-284816d92a15) and accessed the data on 02/21/2025.

6. [1 mark] Write code below to import your data into R. Assign your dataset to an object. Make sure to include and annotate this code in your submission (you can use a # to comment out regular text within code chunks to annotate).

```r
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv",
                 col_types = cols(
                   Year = col_double(),  # Numeric (Discrete)
                   `113 Cause Name` = col_character(),  # Categorical
                   `Cause Name` = col_character(),  # Categorical (Text)
                   State = col_character(),  # Categorical (Text)
                   Deaths = col_double(),  # Numeric (Discrete)
                   `Age-adjusted Death Rate` = col_double()  # Numeric (Continuous)
                 ))
```

7. [3 marks] Write code in R (included in your submission with annotation) to answer the following questions: i) What are the dimensions of the dataset?

```
dimension=dim(data)
dimension
```

```
## [1] 10868     6
```

ii) What are the variable names of the variables in your dataset?

```
variable_name=colnames(data)
variable_name
```

```
## [1] "Year"                          "113 Cause Name"
## [3] "Cause Name"                    "State"
## [5] "Deaths"                        "Age-adjusted Death Rate"
```

iii) Print the first six rows of the dataset.

```r
first_six_row=head(data)
first_six_row
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Accidents (unintention~ Unintention~ Unit~ 169936                   49.4
## 2  2017 Accidents (unintention~ Unintention~ Alab~   2703                   53.8
## 3  2017 Accidents (unintention~ Unintention~ Alas~    436                   63.7
## 4  2017 Accidents (unintention~ Unintention~ Ariz~   4184                   56.2
## 5  2017 Accidents (unintention~ Unintention~ Arka~   1625                   51.8
## 6  2017 Accidents (unintention~ Unintention~ Cali~  13840                   33.2
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

8. [2 marks] Use the data to demonstrate a data visualization skill we have covered during Part I of the course. Choose a visualization relevant to your stated problem. Include your code in your submission. For example, you could visualize the distribution of our outcome with a histogram, or use a bar graph to represent the distribution of your exposure variable.

```
HD_2017 <- data %>% filter(`Cause Name` == "Heart disease", Year == 2017)
```

```
region_mapping <- c(
  'Connecticut' = 'Northeast', 'Maine' = 'Northeast', 'Massachusetts' = 'Northeast',
  'New Hampshire' = 'Northeast', 'Rhode Island' = 'Northeast', 'Vermont' = 'Northeast',
  'New Jersey' = 'Northeast', 'New York' = 'Northeast', 'Pennsylvania' = 'Northeast',
  'Illinois' = 'Midwest', 'Indiana' = 'Midwest', 'Michigan' = 'Midwest',
  'Ohio' = 'Midwest', 'Wisconsin' = 'Midwest', 'Iowa' = 'Midwest',
  'Kansas' = 'Midwest', 'Minnesota' = 'Midwest', 'Missouri' = 'Midwest',
  'Nebraska' = 'Midwest', 'North Dakota' = 'Midwest', 'South Dakota' = 'Midwest',
  'Delaware' = 'South', 'Florida' = 'South', 'Georgia' = 'South',
  'Maryland' = 'South', 'North Carolina' = 'South', 'South Carolina' = 'South',
  'Virginia' = 'South', 'West Virginia' = 'South', 'Alabama' = 'South',
  'Kentucky' = 'South', 'Mississippi' = 'South', 'Tennessee' = 'South',
  'Arkansas' = 'South', 'Louisiana' = 'South', 'Oklahoma' = 'South',
  'Texas' = 'South', 'Arizona' = 'West', 'Colorado' = 'West', 'Idaho' = 'West',
  'Montana' = 'West', 'Nevada' = 'West', 'New Mexico' = 'West', 'Utah' = 'West',
  'Wyoming' = 'West', 'Alaska' = 'West', 'California' = 'West', 'Hawaii' = 'West',
  'Washington' = 'West', 'Oregon' = 'West', 'District of Columbia' = 'South')
```

```
HD_2017 <- HD_2017 %>%
  mutate(Region = region_mapping[State])
```

```
HD_2017 <- HD_2017 %>% filter(Region !="NA")
HD_2017
```

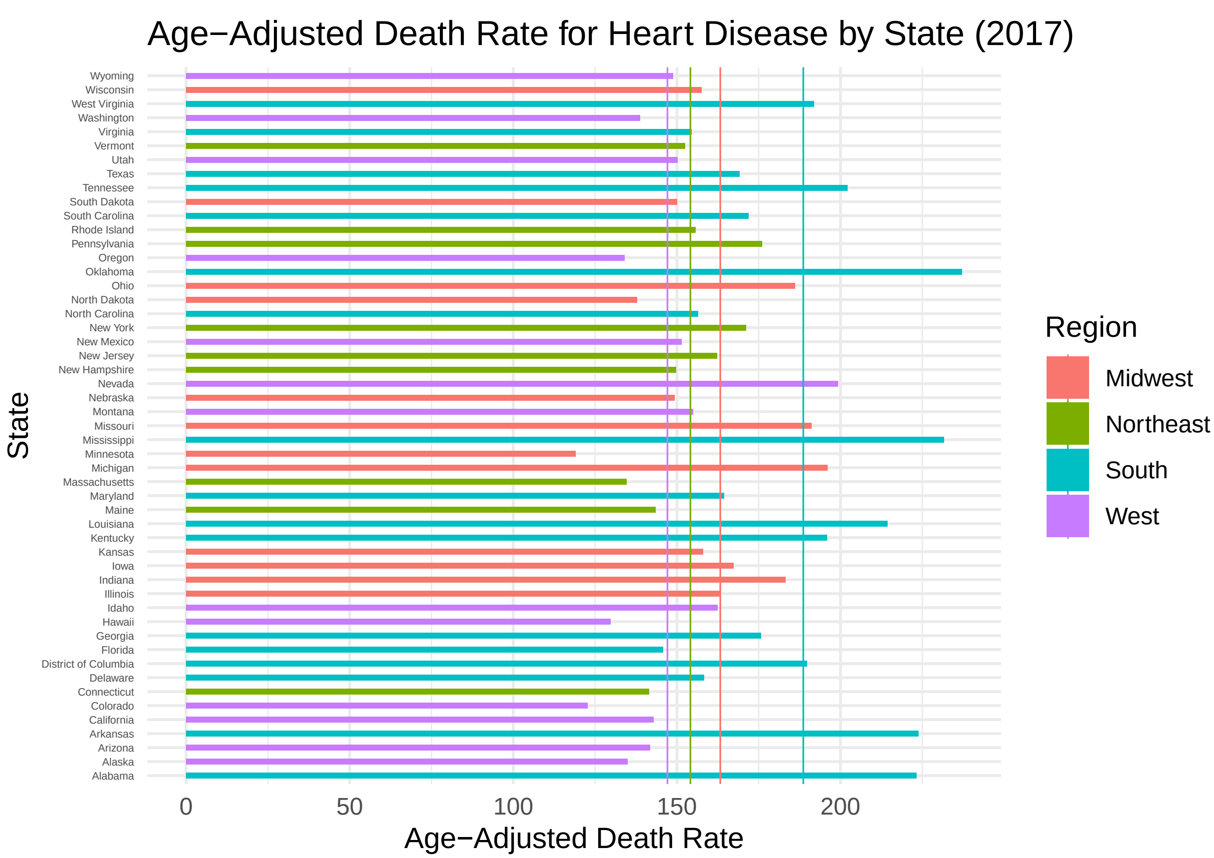
```
## # A tibble: 51 x 7
##     Year `113 Cause Name`       `Cause Name` State Deaths Age-adjusted Death R~1
##    <dbl> <chr>                  <chr>        <chr>  <dbl>                  <dbl>
##  1  2017 Diseases of heart (I0~ Heart disea~ Alab~  13110                   223.
##  2  2017 Diseases of heart (I0~ Heart disea~ Alas~    814                   135
##  3  2017 Diseases of heart (I0~ Heart disea~ Ariz~  12398                   142.
##  4  2017 Diseases of heart (I0~ Heart disea~ Arka~   8270                   224.
##  5  2017 Diseases of heart (I0~ Heart disea~ Cali~  62797                   143.
##  6  2017 Diseases of heart (I0~ Heart disea~ Colo~   7060                   123.
##  7  2017 Diseases of heart (I0~ Heart disea~ Conn~   7138                   142.
##  8  2017 Diseases of heart (I0~ Heart disea~ Dela~   1990                   158.
##  9  2017 Diseases of heart (I0~ Heart disea~ Dist~   1284                   190.
## 10  2017 Diseases of heart (I0~ Heart disea~ Flor~  46440                   146.
## # i 41 more rows
## # i abbreviated name: 1: `Age-adjusted Death Rate`
## # i 1 more variable: Region <chr>
```

```
regional_avg <- HD_2017 %>% group_by(Region) %>% summarize(Average_Rate = mean(
          `Age-adjusted Death Rate`,
          na.rm = TRUE))
regional_avg
```

```
## # A tibble: 4 x 2
##   Region    Average_Rate
##   <chr>            <dbl>
## 1 Midwest           163.
## 2 Northeast         154.
## 3 South             189.
## 4 West              147.
```

```r
plot_HD_2017 <- ggplot(HD_2017, aes(x = `Age-adjusted Death Rate`, y = State, fill = Region)) +
  geom_bar(stat = "identity", width = 0.4) +
  geom_vline(data = regional_avg, aes(xintercept = Average_Rate, color = Region), linewidth = 0.3) +
  labs(title = "Age-Adjusted Death Rate for Heart Disease by State (2017)",
       x = "Age-Adjusted Death Rate",
       y = "State") +
  theme_minimal()+
  theme(axis.text.y = element_text(size = 4))

plot_HD_2017
```



9. [2 marks] Describe the skill that you are demonstrating and interpret your findings. For example, if you have created a histogram, describe the central tendency, shape of the distribution, etc.

- Skill:

1. Data Filtering by using filter(): I used filter(Region != "NA") to ensure that only states with valid region classifications are included in the analysis. This step removes incomplete or irrelevant entries which may cause confusion in interpreting the plot. I also used filter() to filter the original dataset to include only rows where the cause of death is "Heart disease" and the year is 2017.
2. Data Aggregation by using group_by() and summarize():I calculated the average age-adjusted death rate for each region with group_by(Region) and summarize() to get meaningful comparisons between the average regional death rate. By using na.rm = TRUE, this ensures that the averages are accurate by removing missing values.
3. Data Visualization by downloading ggplot2 package: I used ggplot package generally for later plotting.
4. Horizontal Bar Chart by using geom_bar(): I showed each state's age-adjusted death rate with bars being colored by region for an easier comparison.
5. Regional Average Lines by using geom_vline():I added vertical lines for each region's average death rate, allowing for immediate visual assessment of whether a state is above or below its regional average.
6. Minimal Theme by using theme_minimal(): I removed unnecessary grid lines and background elements, making the chart cleaner and easier for interpretation.
7. Data Categorization: I created a mapping of U.S. states to their corresponding geographic regions based on standard regional classifications (e.g., Northeast, Midwest, South, West).
8. Data Transformation by using mutate(): I added a new variable Region to the dataset by mapping the State column to its corresponding region from the region_mapping vector.

- Interpretation:

1. The Southern regions have higher age-adjusted death rates for heart disease because of the longer bars and higher regional average lines.
2. The Western regions generally exhibit lower death rates, mostly falling below their regional average.
3. States like California, Hawaii, and Colorado have significantly lower rates, suggesting potential regional health advantages while Mississippi, Oklahoma, and Alabama show several highest death rates, exceeding both their regional and likely national averages.
4. States above their regional averages like Mississippi in the South may benefit from targeted public health interventions, and States that fall below their regional average like California in the West may serve as case studies for developing better heart disease prevention strategies.